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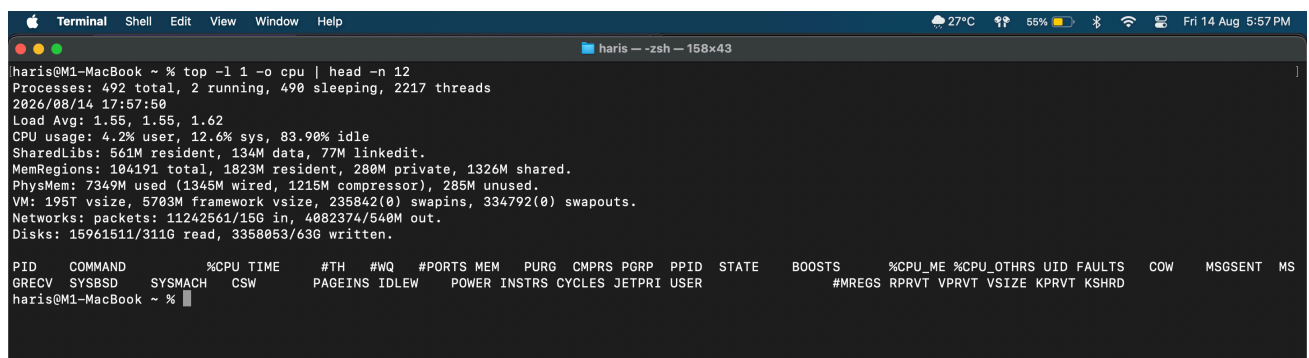
Subject: Linux OS Scripting CYE 5103

Date: 14/08/2026

1. What If There Were No Operating System?

Imagine a computer without an operating system. Identify the problems a user would face and investigate how OS features solve them. Demonstrate at least three features using Linux and present a “Without OS vs With OS” comparison.

Without an OS, a user would face major problems regarding management of the processes and files. Programs couldn't reliably start or run because the computer wouldn't provide standard way to load code, create processes or handle I/O. Every program would have to directly control hardware like disks, keyboard, display, network, making multitasking impossible and causing crashes to frequently take the whole system down. There'd also be no consistent protection between programs, so one misbehaving app could overwrite memory, read other app's data , or lock up the machine. There would also be no unified way of accessing file, permissions, or devices, so even basic tasks like saving documents or using printers would be very chaotic.



```
Terminal Shell Edit View Window Help
haris@M1-MacBook ~ % top -l 1 -o cpu | head -n 12
Processes: 492 total, 2 running, 490 sleeping, 2217 threads
2026/08/14 17:57:50
Load Avg: 1.55, 1.55, 1.62
CPU usage: 4.2% user, 12.6% sys, 83.90% idle
SharedLibs: 561M resident, 134M data, 77M linkedit.
MemRegions: 104191 total, 1823M resident, 280M private, 1326M shared.
PhysMem: 7349M used (1345M wired, 1215M compressor), 285M unused.
VM: 195T vsize, 5703M framework vsize, 235842(0) swapins, 334792(0) swapouts.
Networks: packets: 11242561/15G in, 4082374/640M out.
Disks: 15961511/311G read, 3358053/63G written.

PID    COMMAND    %CPU TIME    #TH    #WQ    #PORTS MEM    PURG    CMPRS    PGRP    PPID    STATE    BOOSTS    %CPU_ME    %CPU_OTHRS    UID    FAULTS    COW    MSGSENT    MS
GRECV  SYSBSD    SYSMACH    CSW    PAGEINS    IDLEW    POWER    INSTRS    CYCLES    JETPRI    USER
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Terminal Shell Edit View Window Help
haris@M1-MacBook ~ % ls -al
total 240
drwxr-xr-x+ 45 haris staff 1440 14 Aug 17:56 .
drwxr-xr-x  6 root  admin  192  9 Aug 20:18 ..
drwxr-xr-x@  3 haris staff   96  2 Oct 2025 .cache
-r-----  1 haris staff    7 15 Sep 2025 .CFUserTextEncoding
drwxr-xr-x@  3 haris staff   96 18 May 11:49 .config
drwxr-xr-x@ 15 haris staff  480 18 Jun 2025 .docker
-rw-r--r--@  1 haris staff 28676 11 Aug 21:23 .DS_Store
drwxr-xr-x@  5 haris staff  160 30 Nov 2024 .eclipse
-rw-r--r--  1 haris staff   97 17 Dec 2024 .gitconfig
-rw-----  1 haris staff   20 11 Aug 21:34 .lessht
drwx-----@  4 haris staff  128 23 May 2025 .mongodb
-rw-----  1 haris staff   39  1 Dec 2024 .mysql_history
-rw-----  1 haris staff   54 19 Apr 2025 .node_repl_history
drwxr-xr-x@  6 haris staff  192  6 Jan 2025 .npm
drwxr-xr-x@  8 haris staff  256 30 Nov 2024 .p2
drwx-----@  9 haris staff  288  6 May 2025 .pgadmin
-rw-r--r--@  1 haris staff   19 19 Feb 2025 .popsql.json
-rw-----  1 haris staff   25  5 May 2025 .psql_history
-rw-----  1 haris staff   18 17 May 13:33 .python_history
drwx-----  7 haris staff  224 17 Jun 2025 .ssh
drwxr-xr-x  4 haris staff  128  2 May 2025 .th-client
drwx-----+ 12 haris staff 384 13 Aug 23:41 .Trash
-rw-----  1 haris staff 18847 17 Jan 2026 .viminfo
drwxr-xr-x@  5 haris staff   96 18 Nov 2024 .vscode
drwxr-xr-x  3 haris staff   96 18 May 20:43 .vscode-shared
-rw-r--r--@  1 haris staff  448 17 Dec 2024 .zprofile
-rw-----  1 haris staff 29041 14 Aug 00:49 .zsh_history
drwx----- 22 haris staff   704 14 Aug 16:53 .zsh_sessions
-rw-r--r--  1 haris staff    0  1 Dec 2024 .zshrc
drwx-----@ 22 haris staff   704  9 Aug 20:33 Applications
drwxrwxrwx  2 haris staff   64 18 Aug 2025 DaVinci Resolve Media
drwx-----+  5 haris staff  160 25 Mar 17:28 Desktop
drwxr-xr-x 11 haris staff  352 29 Jul 22:03 Developer
drwx-----+ 12 haris staff  384 30 Jul 20:18 Documents
drwx-----@ 58 haris staff 1856 11 Aug 14:41 Downloads
drwxr-xr-x@  4 haris staff   128  1 Dec 2024 eclipse
drwx-----+ 103 haris staff 3296 26 May 02:08 Library
drwx-----  6 haris staff  192 30 Nov 2024 Movies
drwx-----+  7 haris staff  224 26 Feb 2025 Music
drwx-----+  9 haris staff  288 13 May 14:39 Pictures
drwxr-xr-x@  3 haris staff   96 30 Nov 2024 Postman

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Terminal Shell Edit View Window Help
haris@M1-MacBook ~ % ls -l /etc/passwd
-rw-r--r--  1 root  wheel  9196  1 Aug 22:01 /etc/passwd
haris@M1-MacBook ~ %

```

Scenario	Without OS	With OS
Running programs	Programs would directly manage hardware and likely fail or conflict	Linux OS loads and manages programs as processes; the kernel handles startup, resources, and scheduling
File handling	No common interface for saving , locating, and organising data in consistent manner.	Linux provides a filesystem (e.g., ext4, XFS) with paths, directories, and atomic file operations.
Protecting system or app data	One buggy program could overwrite memory or interfere with others	Memory protection and per-process isolation prevent many forms of cross-program corruption

2. Imagine your team is setting up a cybersecurity laboratory. Explore at least five Linux distributions relevant to cybersecurity and recommend suitable distributions for different roles such as penetration testing, digital forensics, privacy, and security monitoring. Present your findings creatively as a comparison chart, poster, infographic, or “OS selection guide.”

Linux Distro	Roles	Practicality
Kali Linux	Penetration Testing	Large, curated security toolset. Fast to get started and lot of pentest workflow tools
Parrot Security OS	Privacy, general security and some forensics	Security-focused tooling with privacy friendly options, good for mixed tasks
BlackArch	Power Pentesting	Massive repository of security tools. Highly modular via packages
Whonix	Privacy and anonymity focused environments	Architecture designed for anonymity workflows. Usually with TOR networking. Compartmentalisation of tasks.
CAINE	Digital Forensics	Guided workflows and investigations tooling
TailsOS	High privacy incident handling	Amnesic, privacy first live OS. Reduces persistence on the system
REMnux	Malware analysis and incident traige	Network/malware-focused environment with specialized tooling
OpenWrt	Network monitoring	Runs on routers. Hence good for isolating lab networks and monitoring

3. Exit status:

Execute at least five Linux commands that produce both successful and unsuccessful results. Investigate their exit status using '\$?'.

Create a simple visual/table showing *Command → Result Exit Status → Meaning*.

Command	Result of Exit Status	What it means?
ls -la	0	The command successfully ran
ls *.c (no such file or directory)	2	No such file or directory
grep -e “something” statement.txt	1	No match

somebadcmd	127	No such command
touch ~/somefile/something.txt	1	No such file or directory

Some common Exit codes

Code	Meaning
0	Success
1	Unclassified or general error
2	Incorrect usage of commands
126	Permission denied or no permission to run
127	Command not found
130	Script terminated with CTRL + C (128+2 = 130). SIGINT script termination

What can a script learn from an exit status?

A script can use an exit status to decide what happens after it runs a command, even without parsing an output. By checking the code, it can make decision like retrying, falling back to an alternative command or stopping early. In other words, exit status are simple, reliable control signal for branching logic and error handling.